Supporting Information

An agentic framework for X-ray absorption spectroscopy of catalytic interfaces

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S1 Interface construction and structural labels

The lateral interface construction represents two distinct fcc metals with matching (111) facets. Metal identities and row counts define the ordered stripe geometry. Adsorption regions distinguish boundary rows, nearby rows, and terrace regions. Available placement controls include single-binding adsorbates, multidentate species, and coupled carbon-containing intermediates. The broader Cu–Au sampling plan includes 1:1, 2:1, 1:2, 3:1, and 1:3 stripe ratios as well as pure-metal reference surfaces. These ratios describe the construction space; they do not imply that every parent or adsorbate combination is present in the completed spectral cohort.

Molecular atom correspondence is retained through relaxation. Chemical labels distinguish the requested intermediate from the molecular components in the relaxed calculation structure. Initial molecule mappings are reconciled with the saved atom order, and local metal layers and contacts are refreshed after relaxation. Geometric screening considers short contacts, periodic images, molecular formulas, bonds and angles, unintended coupling between molecules, and placement relative to the surface. The general interface checker uses periodicity in the surface plane and a nonperiodic vacuum direction. Its distance rules are structural screening criteria rather than electronic bond-order or thermodynamic assignments.

The completed-dataset component analysis uses the C/O/H connectivity in the actual structure used for spectral calculation. Pair cutoffs are 1.85, 1.70, 1.30, 1.20, 0.90, and 1.65 Å for C–C, C–O, C–H, O–H, H–H, and O–O, respectively. Verified molecular identities require compatible bond graphs, not elemental composition alone. An OCCO assignment requires a C–C bond, two C–O bonds, no O–O bond, and the corresponding carbon and oxygen degrees. The position of hydrogen distinguishes CHO from COH and COCHO from OCCOH. Assignments must remain unchanged when cutoffs are scaled by 0.9 and 1.1; ambiguous components are retained for review rather than used as identity targets. The shared audit uses in-plane minimum images and a nonperiodic vacuum direction. The reported primary predictions remain those of the archived cohort.

For species s, molecular coverage is N_s/N_{ref}, where N_{ref} is the declared surface reference population. The reference and molecular counts are retained separately from the number of occupied metal sites. For OCCO plus spectator CO, carbon-equivalent coverage is

$$\theta_{\mathrm{C}} = (2N_{\mathrm{OCCO}} + N_{\mathrm{CO}})/N_{\mathrm{ref}}. \tag{S1}$$

Thus two CO molecules and one OCCO molecule can have equal carbon loading but different molecular coverages. Coadsorbed species are represented by simultaneous species-presence labels. COCO is a legacy alias of the bonded OCCO construction template; the unbonded comparator is denoted 2CO. Observed connectivity determines the relaxed-state label.

S2 Electronic-structure and spectral calculations

Structural relaxation is performed with VASP.[1] The committed Cu–Au relaxation examples use a 520 eV plane-wave cutoff, electronic convergence of 10^{-5} eV, a force threshold of 0.05 eV Å^{-1}, fixed cell vectors, and a Γ-centered $1\times1\times1$ mesh. The adjacent-CO example fixes the bottom three metal layers and relaxes the top two. A dipole correction is applied normal to the asymmetric slab. These are input-example parameters; the completed-cohort records retain the available per-calculation settings.

The present inference dataset uses FDMNES Green-function multiple-scattering calculations.[2] Parsed settings for the representative CO configurations include self-consistency, a 6.0 Å cluster radius, the Perdew option, photon-energy output, and a relative energy range of −5 to 50 eV in 0.2 eV steps. The recorded convolved spectra span 8974.0–9029.0 eV with 276 points. Available record fields do not independently specify every core-hole or broadening parameter. The spectra analyzed here are the saved convolved outputs; no new broadening or energy alignment was fitted during model evaluation.

The archived layer inputs contain SCF, Green, and Perdew keywords but no explicit SCFexc, Screening, nohole, or Gamma_hole setting. The output banners identify a 28 July 2026 program revision. The available user guide describes a ground-state self-consistent calculation without a core hole at the K edge, followed by introduction of the hole and screening charge for the absorption calculation.[3] This is an inference from the inputs and documented defaults, not an explicitly recorded core-hole specification; the available guide is dated November 2022. Of the 348 layer spectra, 332 have separate convolution logs reporting Gamma_hole = 1.55 eV and Gamma_max = 10 eV; the other 16 do not have separate convolution logs. These settings therefore cannot be asserted from logs for every layer. A matching guide or source revision is unavailable, so the core-hole prescription is not treated as independently established for that executable. The benchmark reference logs instead identify 11 April 2025 and must not be substituted for the ML-cohort logs.

The framework also generates FEFF and VASP spectroscopy inputs.[4, 5] Shared structural linkage does not imply identical absorber populations or equivalent numerical approximations between backends. The numerical inference scores in the main paper concern FDMNES Cu K-edge data only. Carbon and oxygen edges and other backends form separate potential comparison channels.

S3 Data quality and absorber-population checks

The source inventory contains 172 locations and 155 unique physical structures. Identical source geometries are counted once, including when multiple directories refer to the same structure. Calculation copies within site or layer folders are not counted as new independent configurations. For the full inventory, an unreadable or empty relaxed-structure file can be replaced by a readable initial structure; its presence does not establish a completed relaxation.

A completed spectral record must provide finite, nonconstant Cu K-edge arrays for layers 2–5 on compatible energy grids. Layer ordering is checked against absorber heights, the physical top Cu sites, and the position of adsorbates. Spectral arrays are compared with their saved convolved output, while coordinates, element identities, and atom indices are checked against the relaxed simulation structure. The six FDMNES lattice parameters are reconciled with the relaxed cell before coordinates are compared in a common frame; matching fractional coordinates alone is insufficient. Length and angle tolerances are 2×10^{-5} Å and 2×10^{-5} degrees, respectively. Records with inconsistent chemical identities or unresolved molecular assignments are not accepted as training labels.

The final convolution population is reconciled with the layer absorber table. This is necessary because a completed directory can include both full-layer calculations and later subset calculations. In two CO configurations, the final convolution represents 36 sites although subset main inputs

or intermediate logs describe three sites. Full-population assignment was established by matching the final convolution contributions to their raw spectra, using an energy-cutoff tolerance of approximately 0.00051 eV, complete atom coverage, and compatible grids. These cases are included. A separate CO configuration with an incompatible layer-5 grid is excluded.

Seven records required corrections to molecular identities, counts, or site-species labels. Measurements were unchanged. Duplicate geometries and identical complete spectral stacks were removed before cross-validation. Table S1 summarizes the resulting cohort.

Table S1: Dataset accounting.

Quantity	Count
Source locations	172
Unique inventory structures	155
Completed-record files	102
Duplicate records removed	14
Grid-incompatible record excluded	1
Independent configurations retained	87
Layer spectra retained	348
Mixed-species configurations	9
Single-species configurations	78

S4 Spectral population weighting and feature construction

Let $\mu_l(E)$ denote the raw mean cross section for a layer containing n_l represented Cu sites. Before normalization, the spectrum for a chosen set D of layers is

$$\mu_D(E) = \frac{\sum_{l\in D} n_l \mu_l(E)}{\sum_{l\in D} n_l}. \tag{S2}$$

The seven fixed-depth representations are the four individual layers and averages over the top two, top three, and four available layers. The last representation excludes the unsampled bottom layer and contains no additional attenuation or detection-mode weighting.

For each representation, the minimum intensity is subtracted and the result is normalized by its integrated area. Feature families are: normalized intensity on the recorded energy grid; cumulative trapezoidal integral; first derivative; concatenated smoothed shape, cumulative integral, first derivative, and second derivative; and the first 24 type-II orthonormal discrete-cosine coefficients. Savitzky–Golay smoothing and derivatives use an 11-point window with a third-order polynomial and the energy-grid spacing. The area and transformations are calculated separately for each sample.

Using $\widetilde{\mu}(E)$ for the minimum-subtracted, unit-area-normalized spectrum, the figure labels correspond to

$$\mathrm{Norm}_k = \widetilde{\mu}(E_k), \tag{S3}$$

$$\mathrm{CDF}_k = \int_{E_{\min}}^{E_k} \widetilde{\mu}(E)\,\mathrm{d}E, \tag{S4}$$

$$\mathrm{d1}_k = \left.\frac{\mathrm{d}\widetilde{\mu}}{\mathrm{d}E}\right|_{E_k}. \tag{S5}$$

The numerical derivative uses the stated Savitzky–Golay polynomial fit. Combined concatenates the Savitzky–Golay-smoothed normalized spectrum, CDF, d1, and the second derivative. DCT24 contains coefficients $c_0, \ldots, c_{23}$ of the type-II orthonormal discrete cosine transform of the normalized spectrum; the zero-frequency coefficient is included. The transform follows the type-II cosine

basis.[6] The 24 retained coefficients are a fixed representation dimension in the candidate set, rather than an independently optimized physical cutoff. For N energy samples $x_j = \widetilde{\mu}(E_j)$, the coefficients are

$$c_k = \alpha_k \sum_{j=0}^{N-1} x_j \cos\left[\frac{\pi k(2j+1)}{2N}\right], \qquad \alpha_0 = N^{-1/2}, \quad \alpha_{k>0} = \sqrt{2/N}, \quad k = 0, \ldots, 23. \tag{S6}$$

The representations are alternative numerical views of each spectrum, not additional independent samples.

Additional tree candidates use multiscale spectrum/derivative channels with windows of 7, 11, and 21 points, or the surface spectrum together with a surface-minus-subsurface contrast. The latter channels share the sample's surface-area normalization to preserve relative depth amplitudes. No molecular count, site label, geometry descriptor, or filename is used as a spectral predictor.

S5 Models and nested evaluation

The candidate set includes balanced logistic regression with $C = 1$, an RBF SVM with $C = 3$ and scale-dependent kernel width, multi-output random forests and extra-trees, and independent binary extra-trees for each species.[7] Joint forests use 96 trees; the independent ensembles use 64 trees per species. Random forests use a minimum leaf size of two, while extra-trees use a leaf size of one. Both use square-root feature subsampling. Logistic regression and SVM inputs are standardized using statistics fitted on training data only. A sigmoid of the SVM decision function supplies a thresholdable score rather than a calibrated probability. Constant-label training partitions receive constant scores. PCA is not used in this benchmark.

Five outer folds hold out whole configurations with test sizes of 17, 17, 17, 18, and 18. The original assignments of 85 configurations are retained; two recovered configurations are allocated by label balance and fold size without using spectra or prediction results. Each configuration's four layers and combined-layer variants share the same fold. Three inner folds are drawn within each outer-training partition. Candidate inner assignments are ranked by the number of missing-positive training labels and label imbalance, using training labels alone. Fold assignments are supplied with the analysis outputs.

The primary selector compares independent-species tree representations and thresholds within each outer-training set. The global threshold candidates are 0.15, 0.25, 0.35, 0.45, 0.50, 0.55, 0.65, and 0.75. Guarded species-specific adjustments require at least five positive and ten negative training configurations. Adjustments are restricted to a neighborhood of the global threshold and shrunk halfway toward it; they are retained only when the training-fold selection objective improves. Micro F1 is the primary selection objective. The fixed-depth comparison restricts the candidate set to one depth representation while selecting its features and thresholds.

Micro F1 pools true positives, false positives, and false negatives across species. Macro F1 averages the species-level F1 values over all 11 observed species, including failures on rare labels. Exact identity-set accuracy requires the full predicted species set to match the reference. The independent selector gives micro/macro F1 of 0.779/0.638 and exact-set accuracy of 0.690. Table S2 gives its species-level results.

The fixed extra-trees/first-derivative/surface combination has the highest observed score among the compared fixed entries: micro/macro F1 of 0.798/0.660 and exact-set accuracy of 0.701. This ranking uses outer-fold results and is exploratory. Repeated analysis of the same simulated cohort does not provide an external confirmation set.

Conditional 95% intervals use 1500 bootstrap resamples of held-out configurations, with the fitted predictions held fixed. Paired depth differences use the same resampled configurations. The intervals exclude model-refitting uncertainty and are not corrected for multiple comparisons. The random seed is 20261020.

Table S2: Primary selector by species. Counts are positive configurations and overlap for mixtures.

Species	Positives	Precision	Recall	F1
CHO	5	0.500	0.200	0.286
CO	30	1.000	0.867	0.929
CO2	3	0.667	0.667	0.667
COCHO	1	0.000	0.000	0.000
COH	6	1.000	0.500	0.667
COOH	8	0.714	0.625	0.667
H	22	0.933	0.636	0.757
O	7	1.000	0.714	0.833
OCCOH	4	1.000	0.500	0.667
OCHO	8	0.857	0.750	0.800
OH	5	1.000	0.600	0.750

S6 Figure construction and structural comparisons

All new figure elements are generated by the accompanying plotting code. The circular workflow uses vector graphics, and the substrate rendering in Figure 2a preserves the metal coordinates from the representative input, with adsorbates omitted from that view. The molecular gallery is shown separately, with one element legend shared by the whole panel. Figure 5 preserves the supplied starting coordinates without modifying bond distances. The extended Cu–Au site diagram is an ideal triangular fcc(111) surface with explicitly specified metal identities, adsorbate positions, and site contacts. The Cu–Au bridge has one contact on each side of the indicated boundary. CuCuCu, CuCuAu, and CuAuAu hollows are defined by three explicit metal contacts. Surface and adsorbate coordinates share one oblique projection. Figure 2b assigns each adsorbate-bearing configuration to one single-species or coadsorbed bin; it omits two clean structures and one CuO reference. These schematic coordinates are supplied separately from the calculated input structures. The molecular gallery uses schematic geometries with explicitly declared bond graphs; these illustrations represent chemical connectivity rather than optimized bond lengths. All molecular drawings share the same coordinate scale and display radius for atomic symbols; these symbol sizes do not represent ionic or van der Waals radii.

The reference panels use numeric calculation outputs and experimental arrays processed with reference-v2 (Section S8). The exact fixed-window 0–1 arrays feed both display and similarities; no separate maximum division or earlier edge-jump scores are mixed with them. No graphical display paths are used for metrics. The layer-resolved panel uses recorded CO spectral arrays. The previous species-present/absent descriptive contrast is omitted from the main figure because its reference contains other adsorbates and cannot isolate an adsorbate-induced change. Its historical arrays are retained for provenance only, not as a clean control.

The carbon-coupling figure uses four actual starting models containing two C and two O atoms. Minimum input C–C distances are approximately 2.52 Å for adjacent CO at the Cu-side interface, 2.46 Å for adjacent CO on the Cu terrace, and 1.45 Å for both OCCO models. These distances are not relaxed bond lengths. A matched spectral test requires the same parent, loading, numerical settings, and absorber-population definitions, together with verified final connectivity. No eligible intact-OCCO positives occur in the current inference cohort.

S7 Reproduction and software integration

The archived held-out predictions and reported tables can be replayed without refitting. A separate natural-language-agent run invoked the archived primary selector on a relocated, byte-hashed 87-configuration export and rebuilt 895 fit-cache entries. It reproduced the primary micro/macro F1 of 0.7791/0.6383 and exact identity-set accuracy of 0.6897. Primary and exploratory-winner

prediction arrays matched the archive. Across the broader comparison, 140 of 169 held-out array files matched exactly; 29 differed, predominantly in logistic-regression and support-vector variants, with differences also in some random-forest and macro-selected outputs. The rerun used NumPy 2.4.6 and scikit-learn 1.9.0, compared with archived versions 2.3.5 and 1.8.0. These differences are consistent with dependency sensitivity but do not isolate its cause. The archive was not overwritten. All five newly saved outer-fold models reproduced their held-out predictions after reload. This repeats the original cohort and does not constitute independent confirmation.

The agent and graphical interface call a shared audited-training route. Its archived-primary option is distinct from a separate global-threshold integration recipe. A local execution fixture links simulated scheduler events, a saved relaxed geometry, an existing FDMNES record, and held-out model inference. It tests software handoffs; it does not establish production scheduler execution or generation of new spectra.

The full-cohort export contains the relaxed coordinates, spectral records, absorber tables, calculation inputs and outputs, final convolution evidence, frozen folds, and correction provenance needed to repeat the audit and fitting. Byte hashes protect its link to the archived protocol. Figure-only reproduction uses the accompanying plotted data and requires neither model fitting nor access to the simulation tree. Both modes preserve the distinction between the primary nested selector and the exploratory fixed winner.

The training-most-common-set baseline is fitted separately in each outer-training partition and gives micro/macro F1 of 0.323/0.047 and exact-set accuracy of 0.241. The training-majority binary baseline predicts no species and gives zero for all three measures. All baseline results use the frozen configuration folds.

S8 Reference-v2 processing and similarity evidence

Reference-v1 remains an unchanged archive. The new 3-material by 6-series provenance matrix links each native parsed array to the preserved raw input and SHA-256. Cu, Cu_2O, and CuO formal oxidation labels are not continuous atomic charges. All experimental references are available. Cu_2O PBE and GW and CuO PBE lack usable numeric XAS series after renewed OUTCAR, XML, and HDF5 inspection. Missing cells remain blank. Potential files are not distributed: the local search records only PAW TITEL identifiers, byte counts, and hashes.

S8.1 Native quantities, broadening, and notebook reconciliation

FDMNES mode assignments follow decks/output banners, not directory names. Reference logs identify revision 11 April 2025, radius 7 Å, SCF and Perdew. Five native convolutions report Gamma_hole 1.55 eV, Gamma_max 15 eV, Ecent 30 eV, Elarg 30 eV; these differ from the ML cohort's archived logs. Their energy-dependent broadening is retained with its native conventions. CuO Green includes both saved site contributions; CuO fdm is the raw site-1 curve without completed final-state convolution. FEFF 9.6.4 uses omega and mu, includes its intrinsic lifetime/self-energy (header Gam_ch 1.729 eV), COREHOLE RPA, SCF 7 Å and FMS 9 Å. No second lifetime convolution is applied.

The VASP arrays are trace/3 of the final imaginary dielectric tensor, excluding off-diagonals, for one designated Cu core-hole atom. The native energy is a core-excitation energy requiring photon calibration. Inputs record ICORELEVEL=2, CH_LSPEC, and CH_SIGMA=0.1 eV. The official CH_SIGMA guide identifies Gaussian broadening but does not independently pin the sigma/FWHM interpretation of these executables. The PBE and GW-labelled PAW lineages differ; folder GW does not prove a GW/BSE calculation. Separate Cu calibration offsets are therefore used.

The original notebook uses Gaussian sigma 1 eV and exploratory offsets +120/+125 eV, with another displayed +124 eV route. The actual Cu_GW.dat cell is reproduced and saved separately, but its lineage to the parsed OUTCAR remains unverified. The original Cu_PBE.dat is empty; it is explicitly not substituted while claiming notebook reproduction. The verified OUTCAR PBE is a different identified input. A later cell reuses intensity variables, so a coherent multi-method notebook route cannot be recovered past the empty input. Reference-v2 retains all raw files and these limitations.

The primary is explicitly a native-data relative-intensity comparison: added Gaussian sigma/FWHM and Lorentzian HWHM/FWHM are zero. This removes v1's unsupported universal Lorentzian FWHM 1.55 eV; it does not assert that native CH_SIGMA establishes adequate instrumental broadening. The remaining physical resolution is unknown. A separate additional Gaussian sigma 1 eV (FWHM $2\sqrt{2\ln 2}\sigma = 2.35482$ eV) diagnostic uses a positive normalized kernel. Native lifetime is preserved. No width is fitted. Finite-domain kernel mass is renormalized; uniform-grid interpolation is confined to native support. Raw, native, additionally broadened, and aligned arrays are saved. Unequal native final-state treatments preclude an unconditional accuracy ranking.

S8.2 Background, normalization, and alignment

The first substantive negative calculated absorption arose at v1's local-pre-edge subtraction (not in raw CuO fdm); the latest supplied display reached about -0.209. Native experimental baseline/noise negatives are a separate raw-input issue. The local windows were not established as physical pre-edges for every native energy convention. They are removed in v2, rather than concealed by absolute values, clipping, or a restricted y axis. Raw/pre-normalization diagnostics are retained; min–max scaling is a declared relative-intensity transform, not a repaired physical edge jump.

Processing is native parsing/tensor or duplicate averaging, retained native broadening, optional separately declared positive Gaussian diagnostic, calibrated energy shift, supported interpolation, then the same final min–max rule for every calculation and Experiment. The extrema are evaluated on the identical 8980–9020 eV, 201-point grid. Flat, nonfinite, or insufficient traces fail explicitly. No extrapolation is allowed. Substantive negative calculated pre-normalization absorption fails; a 10^{-12} roundoff tolerance is explicit. No independently justified all-series edge-jump sensitivity is available. This reference normalization does not alter the ML archive's minimum-subtracted unit-area representation.

The sign is $E_{\text{display}} = E_{\text{native}} + \Delta E$. One offset is fitted on Cu for each method and transferred unchanged to both oxides. Bounds (eV) are Green $[-4, 4]$, fdm $[-4, 4]$, FEFF $[-15, -6]$, PBE $[105, 135]$, GW $[110, 145]$, based on photon-output convention, notebook-scale offsets, and distinct native core onsets. Every trial must support the full same comparison window. The objective is min–max RMSE, with a 0.05 eV profile and local bounded refinement. There is one fitted shift per method and no fitted widths. Separately fitted compound shifts within ± 5 eV of the transferable offset are shape diagnostics only. Boundary hits and RMSE+0.01 shift-sensitivity intervals are saved; these are not statistical confidence intervals. Native-axis views are blank where the full photon window is unsupported.

Cosine similarity is the normalized dot product without extra mean centering; Pearson correlates centered intensities; Spearman correlates average ranks with tie handling. Official SciPy cosine, pearsonr, and spearmanr implementations are used. Zero norms, constants, nonfinite arrays, and insufficient overlap yield unavailable metrics rather than zero agreement. Adjacent energy samples are correlated, so per-point correlation p-values are not reported. The identical final arrays feed Figure 3 and the tables. No comparable hardware/resource/problem-size timing logs establish a measured speedup.

Table S3: Complete available reference-v2 similarities. C: Cu-calibrated; S: compound shape alignment; G: added Gaussian diagnostic; N: native axis. All windows are 8980–9020 eV, 201 points. Full units, widths, settings and shifts are also in similarity_metrics.csv.

Material	Method	View	Shift/eV	Cosine	r	ρ
Cu	FDMNES green	C	-0.36	0.998	0.987	0.923
Cu	FDMNES green	S	-0.36	0.998	0.987	0.923
Cu	FDMNES green	N	0.00	0.998	0.986	0.926
Cu	FDMNES green	G	-0.36	0.998	0.987	0.908
Cu	FDMNES fdm	C	1.80	0.994	0.972	0.762
Cu	FDMNES fdm	S	1.80	0.994	0.972	0.762
Cu	FDMNES fdm	N	0.00	0.981	0.883	0.403
Cu	FDMNES fdm	G	1.80	0.993	0.972	0.783
Cu	FEFF	C	-8.52	0.991	0.964	0.910
Cu	FEFF	S	-8.52	0.991	0.964	0.910
Cu	FEFF	G	-8.52	0.992	0.969	0.909
Cu	VASP PBE	C	115.74	0.874	0.428	0.436
Cu	VASP PBE	S	115.74	0.874	0.428	0.436
Cu	VASP PBE	G	115.74	0.946	0.676	0.691
Cu	VASP GW	C	124.78	0.886	0.518	0.621
Cu	VASP GW	S	124.78	0.886	0.518	0.621
Cu	VASP GW	G	124.78	0.977	0.861	0.943
Cu2O	FDMNES green	C	-0.36	0.993	0.975	0.921
Cu2O	FDMNES green	S	0.32	0.997	0.977	0.940
Cu2O	FDMNES green	N	0.00	0.996	0.979	0.937
Cu2O	FDMNES green	G	-0.36	0.995	0.977	0.922
Cu2O	FDMNES fdm	C	1.80	0.988	0.896	0.842
Cu2O	FDMNES fdm	S	-0.29	0.997	0.980	0.989
Cu2O	FDMNES fdm	N	0.00	0.996	0.972	0.993
Cu2O	FDMNES fdm	G	1.80	0.991	0.919	0.843
Cu2O	FEFF	C	-8.52	0.974	0.819	0.587
Cu2O	FEFF	S	-6.00	0.995	0.956	0.948
Cu2O	FEFF	G	-8.52	0.980	0.838	0.597
CuO	FDMNES green	C	-0.36	0.995	0.959	0.916
CuO	FDMNES green	S	0.26	0.997	0.974	0.931
CuO	FDMNES green	N	0.00	0.996	0.969	0.927
CuO	FDMNES green	G	-0.36	0.996	0.969	0.924
CuO	FDMNES fdm	C	1.80	0.962	0.832	0.829
CuO	FDMNES fdm	S	0.55	0.966	0.825	0.766
CuO	FDMNES fdm	N	0.00	0.964	0.795	0.713
CuO	FDMNES fdm	G	1.80	0.976	0.893	0.903
CuO	FEFF	C	-8.52	0.980	0.824	0.785
CuO	FEFF	S	-6.18	0.992	0.950	0.918
CuO	FEFF	G	-8.52	0.982	0.849	0.792
CuO	VASP GW	C	124.78	0.935	0.676	0.679
CuO	VASP GW	S	125.52	0.940	0.730	0.742
CuO	VASP GW	G	124.78	0.955	0.744	0.804

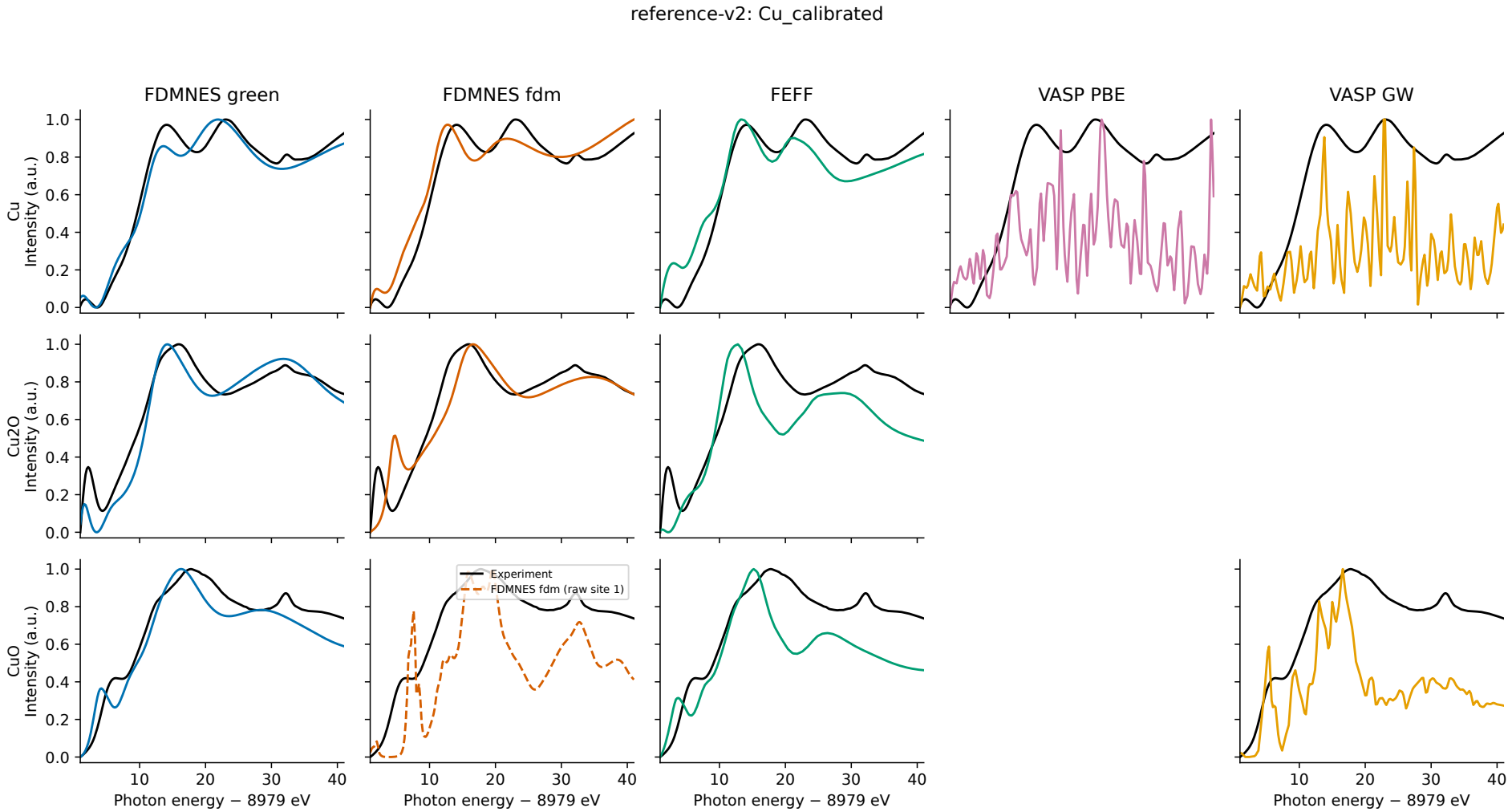


Figure S1: Primary Cu-calibrated native-data comparison. Every trace uses the identical fixed-window min–max rule. Missing calculations are blank. The dashed CuO fdm curve is explicitly keyed as the supplied single site with incomplete native final-state processing.

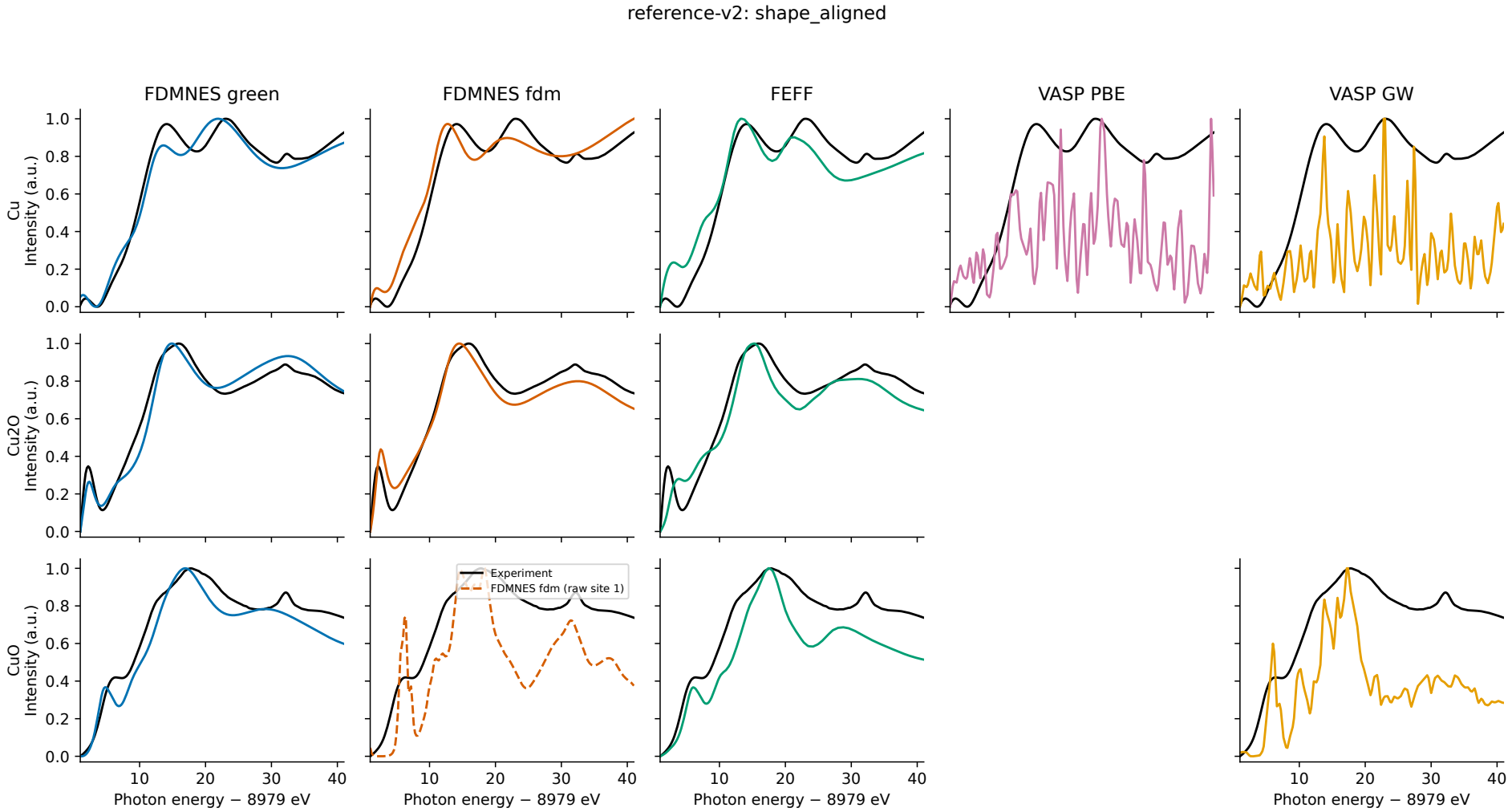


Figure S2: Separately shape-aligned diagnostic: compound-specific shifts remove chemical-shift information.

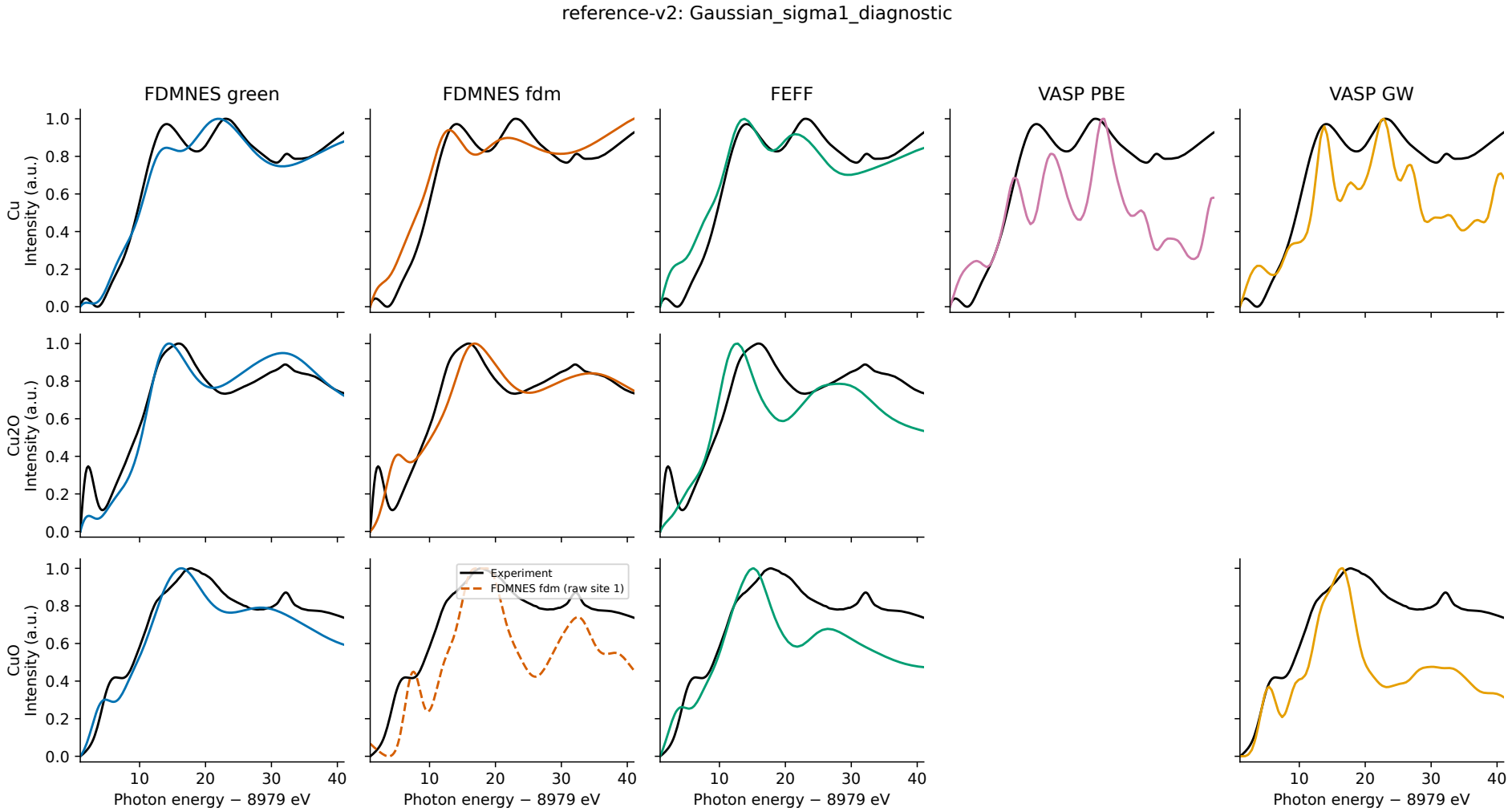


Figure S3: Additional Gaussian sigma 1 eV diagnostic; not measured instrument uncertainty.

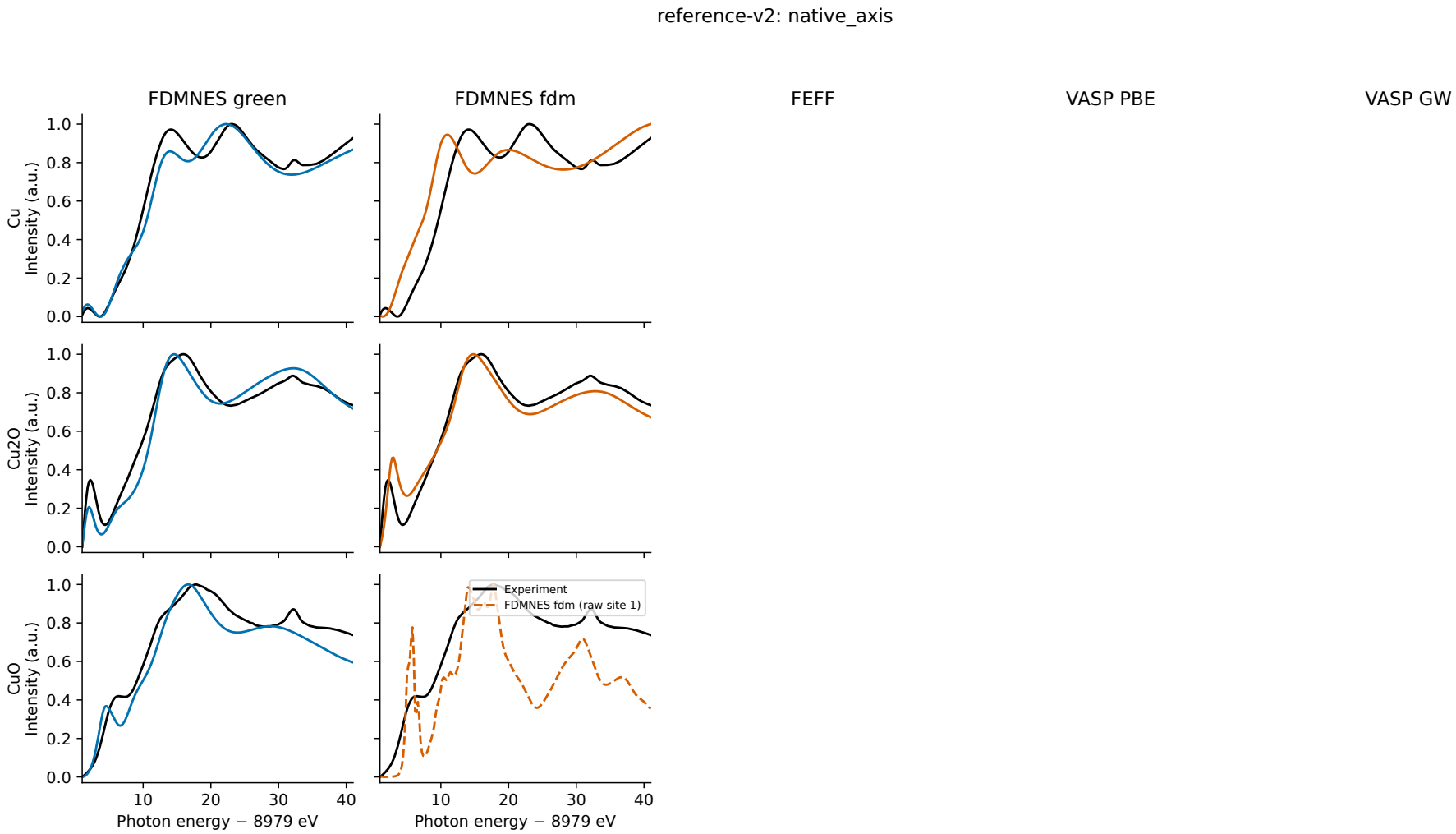


Figure S4: Native axes: insufficient full-window support remains blank rather than extrapolated.

S9 Versioned chemical audit and controlled-output reconciliation

The expanded audit inspected 740 valid geometry files, 2675 artifact paths, and 45 CHO/COH-named or graph-identified aliases, including excluded, incomplete and duplicate structures. Twenty-six candidate aliases are retained ML configurations. Identity uses complete molecular graphs: H bonded to carbon identifies CHO, H bonded to oxygen identifies COH, and disconnected CO/H components remain CO+H. Both in-plane minimum images and the nonperiodic vacuum direction are explicit. Cutoffs are scaled by 0.9 and 1.1; ambiguous outcomes require review. No retained target mismatch or ambiguous retained candidate was demonstrated. The six retained COH directories have CHO basenames but observed O–H connectivity. No targets or frozen results are silently overwritten.

Inventory	CHO only	CHO mixtures	All CHO	All COH
Retained 87	2	3	5	6
Unique 155	2	11	13	15

Table S4: Existing configuration labels, cross-checked with the geometry audit where available. These are configurations, not molecule counts. Single species can mean multiple molecules. Inventory labels are not universal independent validation of every incomplete structure.

The retained CHO mixtures are the interface-Cu, interior-Cu, and uniform CHO 0.056/0.111 entries with CHO;CO;H labels; CHO-only entries are the AuCu- and CuCu-bidentate 0.056 structures. Per-molecule and configuration CSV/JSON tables retain paths, hashes, C–H/O–H distances, bonds, metal contacts, interface distance, intended/relaxed/XAS-used graphs, targets, and inclusion reasons. All available full-cell XAS decks and geometry copies are compared with relaxed coordinates. Element-wise minimum-distance atom assignment detects exact copies and reordered indices, but is not a trajectory-based proof of hydrogen identity exchange in changed geometries. Enlarged local figures display H with a dark border, numbered atoms and explicit bonds; the saved code operates on the frozen audit arrays.

State	Region	Expanded-search outcome
2CO	Interface	Attempt located; no valid relaxed output or verified spectrum for the pinned input.
2CO	Terrace	Completed usable record recovered, with intact two-CO connectivity; initial input differs from pinned quartet. Requires lineage reconciliation or regenerated pinned version before this controlled contrast.
OCCO	Interface	Pinned relaxed geometry and matched Cu spectrum not located; never-run status is not established.
OCCO	Terrace	Pinned relaxed geometry and matched Cu spectrum not located; never-run status is not established.

Table S5: Quartet status derives from recovered artifacts. Equivalent duplicate relaxed geometries count as one outcome. C/O edges are extensions, not prerequisites for the first valid Cu contrast.

Among 102 record attempts, 101 passed geometry/grid/absorber checks and one incompatible grid remains excluded. Seven equal-formula hydrogenation candidates were screened against the same initial metal parent, equal elemental loading, numerical controls and absorber populations, initial in-plane C/O pattern RMS at most 0.75 Å, and relaxed interface-distance mismatch at most 0.5 Å. None passes all checks. Thresholds and each rejection are explicit; equal formula is insufficient. No intended intact-OCCO spectrum is inferred from a changed graph. An intact OCCO-plus-four-CO relaxed geometry is a different loading and has no completed spectrum in the search.

The clean search starts at plain/CONTCAR and plain/2fixed/CONTCAR and also examines all saved XAS geometry copies and raw convolved files. Both clean relaxed structures exist, but no compatible numeric clean spectrum was located. The absence of an AI-ready record alone is not

the criterion. Consequently no adsorbate-minus-clean or coupling difference is calculated. Such a difference must retain a shared per-absorber scale and compatible parent, layers, settings, polarization and energy convention; independently min–max normalized absorption cannot be subtracted as a physical contrast. Separately relaxed and frozen-metal controls have different meanings. Prepared clean full-cell decks and quartet relaxation inputs contain no licensed potentials and were not submitted. Exact missing paths and hashes accompany the status CSV.

S10 Nested ML extensions and their limits

The primary frozen 87-configuration/348-spectrum archive is unchanged. H recall is 14/14 in single-species configurations and 0/8 in mixtures; CO is 21/21 and 5/9. These counts reproduce the supplied cross-check. The exploratory fixed winner is separate. No new fit on these repeatedly examined folds is independent confirmation.

All new recipes use the frozen outer folds and three inner grouped folds. Spectra alone enter predictors; filenames, species/counts, graph/site/coverage/interface descriptors and parent strings define audits, targets or groups only. The surface-layer spectra are cropped to 8976–9027 eV so every predeclared shift of up to 2 eV has native support without extrapolation. The archived full-grid primary is an explicitly different frozen comparator. Existing normalized spectra, CDF and derivatives are included as controls rather than called new. New descriptor windows are fixed at 8976–8985, 8985–8998, 8998–9012 and 9012–9027 eV from edge-relative assumptions, before inspecting test scores; Figure 3's label-conditioned curves do not choose them. Each window yields area, centroid, derivative mean/max/min, peak energy and height. DCT 8/16/24/32 and PCA 8/16 are candidates; PCA/scalers are fitted on training partitions alone. DCT24 retains its historical fixed-candidate meaning.

Nearest neighbors tests cosine/Euclidean and k=1/3/5 inside training, with closest training examples returned for review. Independent Extra Trees and a sequential probability-chain variant each test two compact regularized configurations (32 trees, minimum leaf 2 or 4). Chain training uses group-out-of-fold independent predecessor proxy probabilities; inference proceeds sequentially without true labels. This is explicitly a proxy-chain variant, not an unrestricted co-occurrence generalization claim. Canonical/training-prevalence orders are considered only inside training. Independent per-target XGBoost tests 48/80 rounds, depths 2/3, learning rate 0.05, regularization 5/10, alpha 0.1, minimum child weight 3, row/column fractions 0.8, and training-class weights capped at 10. There is no early stopping against outer folds. Single-class partitions use constant predictors. The saved protocol gives every candidate and missing-positive partition.

Platt calibration uses inner grouped OOF scores only and is unsupported with fewer than five positives or negatives or constant scores. Threshold candidates are 0.15, 0.25, 0.35, 0.45, 0.50, 0.55, 0.65 and 0.75; the global micro-F1 threshold and supported per-label refinements use inner data only. Feature and hyperparameter winners likewise use inner scores. The compact search remains exploratory on repeatedly examined configurations.

None exceeds the primary overall micro F1. Nearest neighbors recovers H 3/8 and CO 6/9 mixtures, with lower overall micro F1; H false positives rise from one to four and CO false positives from zero to two; single-species CO recall falls from 21/21 to 19/21. Per-species precision is reported below rather than treating mixture recall alone as improvement. Paired configuration bootstrap intervals use 500 resamples of fixed OOF predictions; nearest-neighbor micro-F1 change is -0.046 (conditional 95% interval $[-0.139, 0.051]$). These intervals omit refitting/selection variation and do not establish superiority. Refit learning curves separately use fractions 0.25/0.5/0.75/1 and seeds 11/23/47 for a predeclared DCT16 tree recipe with threshold 0.5.

Synthetic stress tests are shifts $\pm$0.2/0.5/1/2 eV, extra Gaussian sigma 0.25/0.5/1 eV, and noise 1/2/5% of native peak intensity. They are scenarios, not measured instrument uncertainty. Positive finite-domain-normalized Gaussian kernels preserve native lifetime treatment. Training-only aug-

Table S6: Frozen primary and new nested extensions. Aug. denotes training-only synthetic augmentation. New recipes use the common cropped window; these are repeated evaluations of the same configurations.

Recipe	Micro F1	Macro F1	Exact set
frozen primary	0.779	0.638	0.690
nearest neighbor	0.733	0.607	0.690
nearest neighbor + aug.	0.714	0.567	0.701
independent trees	0.650	0.476	0.517
independent trees + aug.	0.674	0.502	0.552
classifier chain	0.671	0.505	0.563
classifier chain + aug.	0.663	0.476	0.552
xgboost	0.614	0.440	0.425
xgboost + aug.	0.596	0.371	0.494

mentation supplies one additional combined random ±0.5 eV shift, sigma 0.5 eV Gaussian and 2% noise copy per training configuration. All copies stay in that configuration's fold. Seed 20261022 and perturbations are frozen in the protocol; stress scores do not select models.

The harder evaluation holds out 56 exact initial-metal coordinate groups across three folds, using a predeclared DCT16 independent-tree recipe with training-only thresholds. Its micro/macro F1 and exact-set accuracy are 0.620/0.470/0.552. These coordinate groups are not verified independent ancestral parents; actual parent transfer cannot be estimated from unverified ancestry. Missing-positive labels are reported per split. No generalization to experimental spectra, new compositions or applied potentials is established.

Review curves use minimum calibrated threshold margin among supported labels, excluding predicted unsupported positives; sparse-label uncertainty remains unresolved and no certified full-set risk is claimed. Predictions expose the identity set, scores, thresholds, unsupported calibration and energy/depth/parent domain flags. A local natural-language agent request calls the shared audited-training API, fits the declared recipe, saves/reloads all 40 fold models, predicts held-out records and saves a sanitized review report. All round trips match exactly. This uses available records; no scheduler or new-spectrum execution is implied. Proposed next actions distinguish scientific controls (clean/coupling, matched CO/H, independent hydrogenated C1/C2 with a locked confirmation set) from disagreement review of existing cases. Proposed calculations have not improved a measured result.

Table S7: Per-species extension precision (P), recall (R), F1 and independent configuration support. Aug. is training-only augmentation.

Recipe	Species	P	R	F1	Support
nearest neighbor	CHO	1.000	0.600	0.750	5
nearest neighbor	CO	0.926	0.833	0.877	30
nearest neighbor	CO2	0.400	0.667	0.500	3
nearest neighbor	COCHO	0.000	0.000	0.000	1
nearest neighbor	COH	0.429	0.500	0.462	6
nearest neighbor	COOH	0.667	0.750	0.706	8
nearest neighbor	H	0.800	0.727	0.762	22
nearest neighbor	O	0.667	0.857	0.750	7
nearest neighbor	OCCOH	0.500	0.500	0.500	4
nearest neighbor	OCHO	0.800	0.500	0.615	8
nearest neighbor	OH	1.000	0.600	0.750	5
nearest neighbor + aug.	CHO	0.333	0.200	0.250	5
nearest neighbor + aug.	CO	0.960	0.800	0.873	30
nearest neighbor + aug.	CO2	0.400	0.667	0.500	3
nearest neighbor + aug.	COCHO	0.000	0.000	0.000	1

Recipe	Species	P	R	F1	Support
nearest neighbor + aug.	COH	0.750	0.500	0.600	6
nearest neighbor + aug.	COOH	0.714	0.625	0.667	8
nearest neighbor + aug.	H	1.000	0.682	0.811	22
nearest neighbor + aug.	O	0.667	0.857	0.750	7
nearest neighbor + aug.	OCCOH	0.667	0.500	0.571	4
nearest neighbor + aug.	OCHO	0.444	0.500	0.471	8
nearest neighbor + aug.	OH	1.000	0.600	0.750	5
independent trees	CHO	1.000	0.200	0.333	5
independent trees	CO	1.000	0.833	0.909	30
independent trees	CO2	0.500	0.667	0.571	3
independent trees	COCHO	0.000	0.000	0.000	1
independent trees	COH	1.000	0.333	0.500	6
independent trees	COOH	0.286	0.250	0.267	8
independent trees	H	0.923	0.545	0.686	22
independent trees	O	1.000	0.143	0.250	7
independent trees	OCCOH	0.667	0.500	0.571	4
independent trees	OCHO	1.000	0.250	0.400	8
independent trees	OH	1.000	0.600	0.750	5
independent trees + aug.	CHO	0.000	0.000	0.000	5
independent trees + aug.	CO	0.926	0.833	0.877	30
independent trees + aug.	CO2	0.286	0.667	0.400	3
independent trees + aug.	COCHO	0.000	0.000	0.000	1
independent trees + aug.	COH	1.000	0.500	0.667	6
independent trees + aug.	COOH	0.571	0.500	0.533	8
independent trees + aug.	H	0.824	0.636	0.718	22
independent trees + aug.	O	1.000	0.429	0.600	7
independent trees + aug.	OCCOH	0.667	0.500	0.571	4
independent trees + aug.	OCHO	1.000	0.250	0.400	8
independent trees + aug.	OH	1.000	0.600	0.750	5
classifier chain	CHO	1.000	0.200	0.333	5
classifier chain	CO	0.962	0.833	0.893	30
classifier chain	CO2	0.400	0.667	0.500	3
classifier chain	COCHO	0.000	0.000	0.000	1
classifier chain	COH	1.000	0.333	0.500	6
classifier chain	COOH	0.600	0.375	0.462	8
classifier chain	H	0.929	0.591	0.722	22
classifier chain	O	0.500	0.286	0.364	7
classifier chain	OCCOH	1.000	0.500	0.667	4
classifier chain	OCHO	0.667	0.250	0.364	8
classifier chain	OH	1.000	0.600	0.750	5
classifier chain + aug.	CHO	0.000	0.000	0.000	5
classifier chain + aug.	CO	0.960	0.800	0.873	30
classifier chain + aug.	CO2	0.400	0.667	0.500	3
classifier chain + aug.	COCHO	0.000	0.000	0.000	1
classifier chain + aug.	COH	1.000	0.500	0.667	6
classifier chain + aug.	COOH	0.571	0.500	0.533	8
classifier chain + aug.	H	0.833	0.682	0.750	22
classifier chain + aug.	O	0.000	0.000	0.000	7
classifier chain + aug.	OCCOH	1.000	0.750	0.857	4
classifier chain + aug.	OCHO	0.400	0.250	0.308	8
classifier chain + aug.	OH	1.000	0.600	0.750	5
xgboost	CHO	0.500	0.600	0.545	5
xgboost	CO	0.806	0.833	0.820	30
xgboost	CO2	0.333	0.667	0.444	3
xgboost	COCHO	0.000	0.000	0.000	1
xgboost	COH	0.750	0.500	0.600	6
xgboost	COOH	0.571	0.500	0.533	8
xgboost	H	1.000	0.636	0.778	22

Recipe	Species	P	R	F1	Support
xgboost	O	0.000	0.000	0.000	7
xgboost	OCCOH	0.333	0.500	0.400	4
xgboost	OCHO	0.333	0.125	0.182	8
xgboost	OH	0.400	0.800	0.533	5
xgboost + aug.	CHO	0.333	0.200	0.250	5
xgboost + aug.	CO	0.958	0.767	0.852	30
xgboost + aug.	CO2	0.333	0.333	0.333	3
xgboost + aug.	COCHO	0.000	0.000	0.000	1
xgboost + aug.	COH	0.000	0.000	0.000	6
xgboost + aug.	COOH	0.364	0.500	0.421	8
xgboost + aug.	H	0.824	0.636	0.718	22
xgboost + aug.	O	0.000	0.000	0.000	7
xgboost + aug.	OCCOH	0.400	0.500	0.444	4
xgboost + aug.	OCHO	1.000	0.250	0.400	8
xgboost + aug.	OH	0.571	0.800	0.667	5

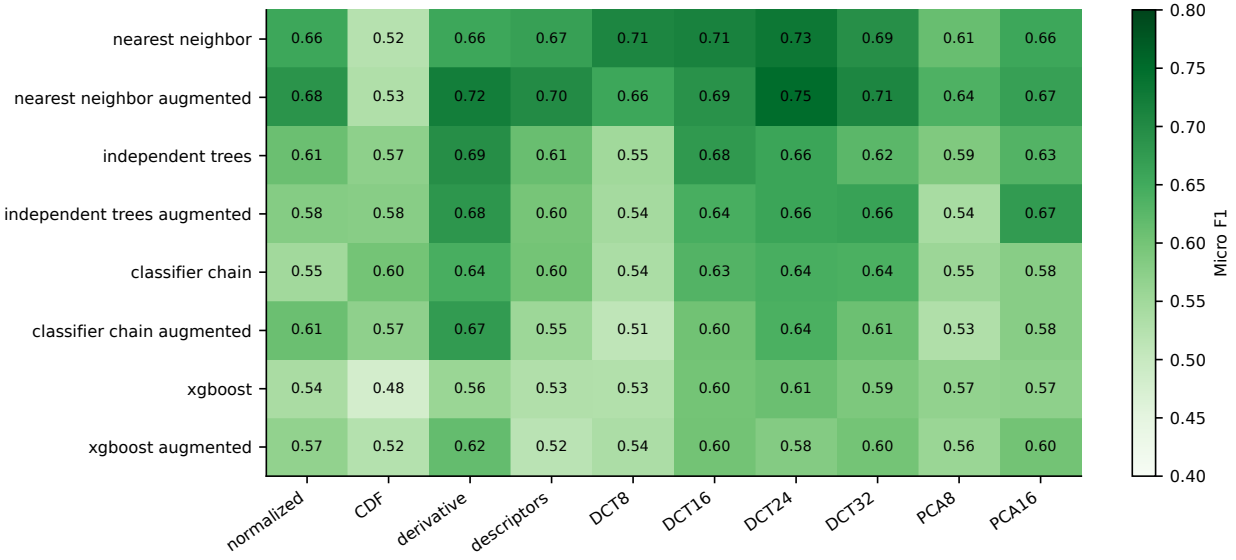


Figure S5: Fixed-feature nested ablations. Each cell uses training-only tuning, not outer-score feature selection.

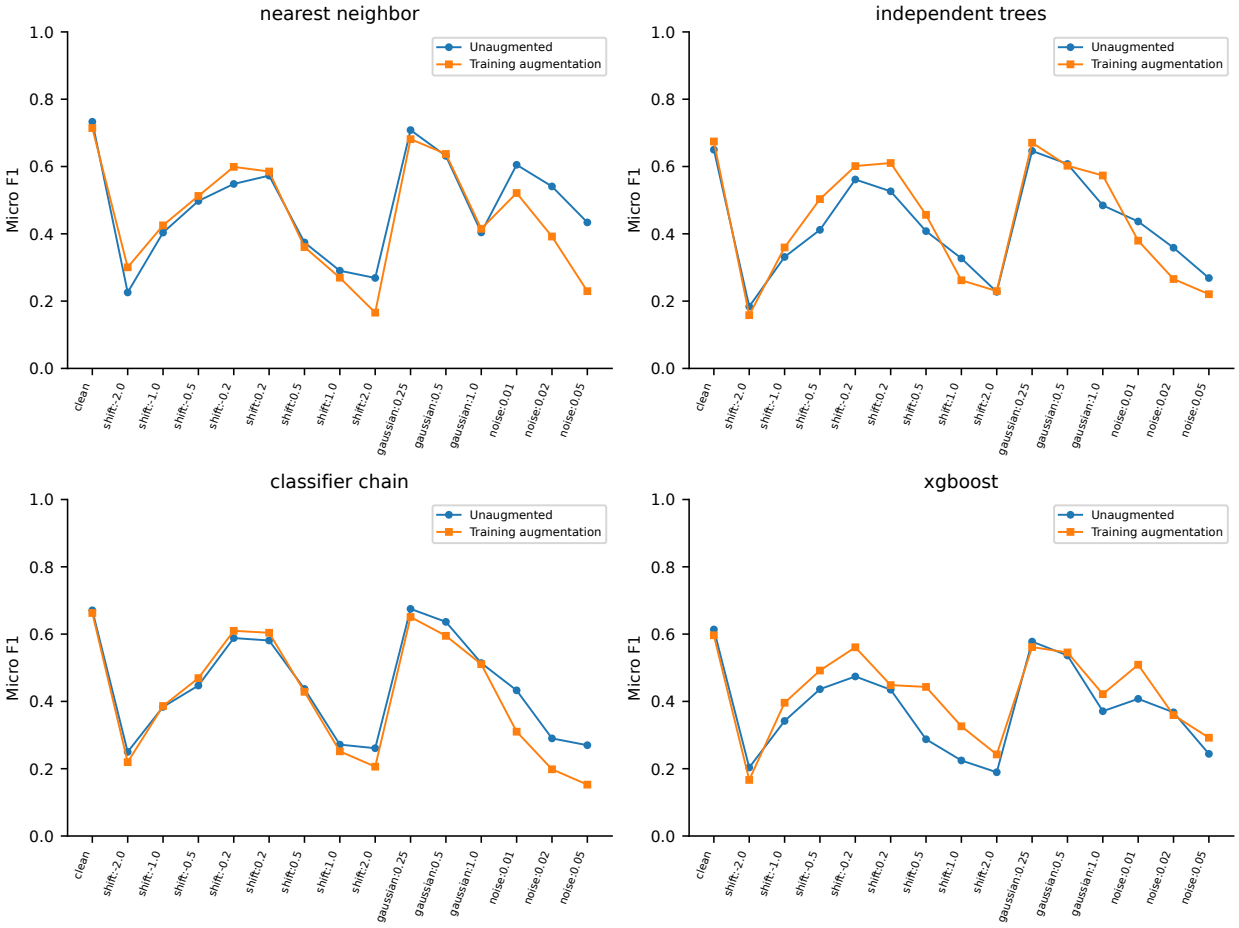


Figure S6: Clean and synthetic held-out stress scenarios for unaugmented and training-augmented recipes. These are not measured uncertainty distributions.

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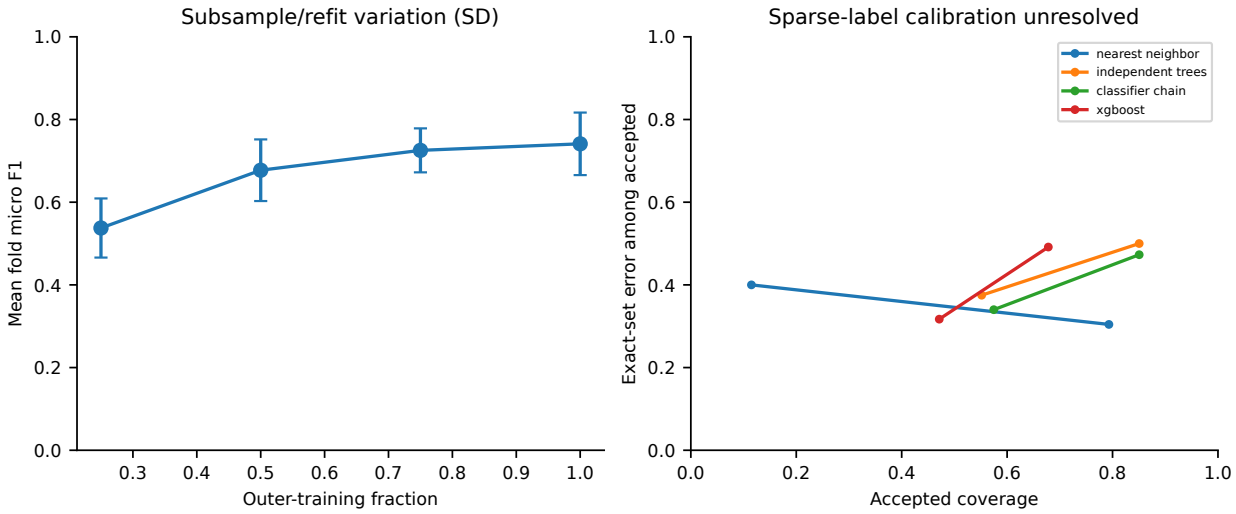


Figure S7: Left: fold/subsample/refit mean and standard deviation, not a prospective confidence interval. Right: coverage versus exact-set error under the review rule; unsupported sparse labels prevent certified full-set calibration.